MUKUL SHARMA

Senior Staff Engineer · Agentic software factories · Ships 1.0 products

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Sixteen years taking products from first commit to first release: SOASTA TouchTest and the mPulse Mobile SDK, Speak Music's Muse, MobileIron AppStation, Adobe Express video, and the Adobe Firefly Video Editor timeline from its first commit to public beta and 100K monthly users. Now building agentic software factories: Architect, an agentic harness 16+ engineers have run nearly 2,000 times, which wrote the code for 400+ of my merged PRs in six weeks.

EXPERIENCE

Sr. Computer Scientist 2 (Sr. Staff Engineer), Adobe, San Jose, CA *Aug 2021 – Present*

- Lead all 22 software and quality engineers on my team as the senior IC for 3.5 years; reviewed about 2,300 PRs and mentored 10 engineers.

Agentic software factory *2026*

- Built Architect v1, an agentic harness that turns a problem statement into a reviewed, CI-green PR: adversarial planning, code, tests, mutation hardening, and ship, gated on human review. 16+ engineers, about 2,000 runs.
- Shipped 400+ merged PRs in six weeks to Adobe Firefly's platform repo, with the production code written by the pipeline; 787 merged since April 2026.
- Pinned agent and prompt behavior with FAIL-to-PASS evals, mutation-probed and gated in CI, plus a layer-aware coverage policy, so prompt edits can't silently regress.
- Wrote the skills and rules for AI-guided development in Adobe's video and Firefly codebases, taught engineers to use them, and helped other teams, including Lightroom and Firefly Boards, adopt agentic development.
- Open-sourced Forge (adversarial multi-agent planning plus a step-by-step execution pipeline), ProofGate (agent reliability gates, built from an audit of 5,020 real sessions), Assistant (an orchestrator for parallel agents), and Perch (drive coding agents from Slack).

Adobe Firefly Video Editor and Firefly video Assembler *2023 – Present*

- Built the timeline editing core, from multi-track and freeform tracks to trim, move, copy and paste, and keyboard shortcuts, from the first commit through public beta (Dec 2025) and 100K monthly active users.
- Led cross-team work on WebCodecs audio, AV sync, and frame decoding that cleared Public Beta blockers, and stabilized the first-mile import and editing flows.
- Contributed the time coordinate system for frame-accurate editing and frame-rate support matching Premiere Pro; contributed to choreography caching that sped up Move by 70% on a 10-track timeline.
- Leading delivery of Firefly video Assembler to public beta in November 2026.

Adobe Express and Project Violet (iOS) *2021 – 2023*

- Led the team that built 8 Video Quick Actions for Adobe Express GA, and resolved several GA blockers.
- Led the team that delivered Mobile Sceneline V1 for Adobe Express iOS.
- Built Dynamic Text, log export, a Project Editor refactor for parallel animations, and the Soundtrack Selector for Project Violet, a SwiftUI video app.

Staff Software Engineer (iOS), MobileIron (acquired by Ivanti), Mountain View, CA *Oct 2018 – Aug 2021*

- Built and shipped AppStation 1.0 for iOS (Jun 2019), MobileIron's app for managing work apps without managing the whole device.
- Lead engineer for MobileIron's enterprise iOS apps, including Mobile@Work and MobileIron Go, in Swift and Objective-C; owned the release lifecycle and app container security.

iOS Platform Lead, Speak Music, San Jose, CA *Jan 2017 – Oct 2018*

- Led design, development, and launch of Muse, an Alexa device for the car (launched Nov 2017; over $70,000 from 1,000+ Indiegogo backers), and owned its iOS app from start to delivery.
- Took over the Melody iOS app, a voice music assistant for iHome speakers, on joining.

Senior Mobile Engineer, 8x8, San Jose, CA *Dec 2015 – Jan 2017*

- Lead developer for the 8x8 Virtual Office iOS app; set up its iOS and Android test automation with TouchTest and Jenkins.

Sr. Software Engineer, then Tech Lead, SOASTA, Mountain View, CA *May 2011 – Dec 2015*

- On the launch team for SOASTA TouchTest (2012), mobile test automation for iOS and Android.
- Architect for mPulse Mobile: designed and built the SDK, API, communication model, and network-request analytics.
- Built dynamic instrumentation that injects SOASTA libraries into IPA and APK files, cutting customers' time to first test; promoted to Tech Lead in 2015.

Software Engineer, SumTotal Systems, Gainesville, FL *Jun 2010 – May 2011*

- Led OpenScript development for version 11 of the ToolBook authoring suite.

SKILLS

AI and agents Multi-agent harnesses, evals and LLM judges, MCP, tool calling, Claude Code, RAG

Languages TypeScript, JavaScript, Python, Swift, Objective-C, Java

Platforms React, Node.js, iOS, Android, WebCodecs, Playwright, Jenkins, Git

PATENTS AND PUBLICATIONS

Load test charts with standard deviation and percentile statistics · *US Patent 9,251,035, issued Feb 2016*

Cloud-based custom metric/timer definitions and real-time analytics of mobile apps · *US 2015/0067527, published Mar 2015*

SHIELD: Social Sensing and Help in Emergency using Mobile Devices · *G. S. Thakur, M. Sharma, A. Helmy, IEEE Globecom 2010*

EDUCATION

M.S. in Computer Science, University of Florida (GPA 3.90 / 4.00) *May 2010*